Doc. 308 – The Cosmic Sector of FFGFT: Redshift, Dark Sector and Geometric Gravity

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What this is about

FFGFT advances an unusual claim: the universe is not expanding. Cosmic redshift is not a Doppler effect but a geometric property of the ξ-field. Dark energy does not exist as a physical entity. And dark matter requires no particles — its effects are to follow from the same geometry that carries light deflection; this is established for gas-dominated dwarf galaxies, for spiral galaxies and the Bullet Cluster it is open.

This document assembles the cosmic sector of FFGFT: what has been derived, what has been observationally confirmed, and where the limits lie. It addresses readers with a physics background who want to know *what* the theory says and *why*.

The three sources of degeneracy are named, the two cores of the geometry are developed, and the status of each step — derivation, structural argument, posit — is declared.

Conventions

Legend.

[K]	*Core derivation* — derived mathematically or numerically from the foundational posits; reproducible.
[S]	*Structural claim* — justified but not yet fully proved.
[POSIT]	*Posit* — declared assumption, not further derived.

Central quantities.

$\xi = 4/30000$	Only fundamental parameter of FFGFT [POSIT, P33].
$\tilde{T} \cdot m = 1$	Foundational relation: intrinsic time and mass are inversely coupled.
$R_H = c/H_0$	Geometric maximal scale — not an expansion horizon.
$a_0 = c^2\xi^{10}/(4\bar{\lambda}_e)$	Transition scale of the inertia regime; below a_0 inertia is non-linear ($\tilde{T} \cdot m = 1$ + Unruh); $= 1.033 \times 10^{-10}$ m/s^2.
$\bar{\lambda}_e = \hbar/(m_e c)$	Reduced Compton wavelength of the electron.

Chapter A

The Static Universe and Redshift

A.1 The fundamental choice

ΛCDM describes an expanding universe: galaxies move apart, light is redshifted by the expansion of space, and the observation of distant supernovae reveals an accelerated expansion accounted for by a cosmological constant Λ.

FFGFT makes a different fundamental choice: the universe is static. Cosmic redshift arises not from motion but in the mass reading: clocks and rulers run with $1/m$, and the geometric path elongation in the ξ-field describes the same redshift within this reading. As an energy loss of the photon alone it is ruled out (Doc. 392).

A.2 The redshift mechanism [K]

The FFGFT redshift formula (Doc. 280):

$$1 + z = e^{\xi x} \qquad \Longrightarrow \qquad z \approx \xi x \quad \text{(for small } z\text{)},$$

where x is the path length, measured in a length that is not derived from ξ and is fixed only via the measured H_0: the form of the relation is derived, its scale is calibrated (P39). The redshift grows with the distance travelled, without wavelength dependence. It is read in the mass reading, in which clocks and rulers run along; only this reading passes the supernova light curves, the background temperature and the Tolman test, while an energy loss of the photon alone fails all three **[X]** (Doc. 392). With x as the coordinate distance of the mass reading, $H(z) = H_0(1 + z)$ follows; read this way the law carries the DESI data up to $z \approx 1.5$ and misses the Lyman-α point at $z = 2.33$ by $5.3\,\sigma$ **[X]** (Doc. 392). A different identification of x has not been tested **[S]**.

Achromatic [K]: The path elongation is purely geometric and stretches *all* wavelengths by the same factor $e^{\xi x}$. Finite-element simulations of the geometry confirm this (K6, Doc. 190). A chromatic redshift would destroy the coherence of the CMB spectral shape — this is not observed.

A.3 The degeneracy of the data [K]

For the three classical pillars of cosmology — Type Ia supernovae, baryon acoustic oscillations (BAO), CMB temperature — the following holds: they cannot in principle distinguish between expansion and static geometric path stretching, as long as both models produce the same distance-redshift relation $z(d)$ (Doc. 267).

This is not a weakness of FFGFT. It is a methodological statement: *these data alone do not decide between the models.* The decision shifts to theoretical consistency, parameter count and derivability.

A.4 What follows from this

For dark energy: Λ is needed in ΛCDM to account for the apparent accelerated expansion. In the static reading there is no expansion to be accelerated. Λ loses its referent in the static reading — it is not falsified, it becomes objectless.

For the Hubble tension: Different measurement methods for H_0 give different values. In the FFGFT reading, H_0 does not measure an expansion rate but a parameter of the ξ-geometry. FFGFT knows only *one*, calibrated H_0 value and does not distinguish between early and late measurements; the tension cannot be modelled in FFGFT — it is neither reproduced nor resolved.

For the z-axis: Every presentation of cosmological data "over time" is formulated in ΛCDM language. An FFGFT interpretation must treat z as a geometric distance parameter, not a time axis.

Chapter B

Geometric Gravity: the Two-Halves Synthesis

B.1 Einstein 1911 and 1915

Einstein calculated the deflection of light at the Sun in 1911 from time dilation alone:

$$\alpha_{1911} = \frac{2GM}{c^2 b} = 0.875''.$$

That was half the measured value. The 1915 theory of GR yielded the full value $\alpha_{1915} = 1.750''$, because space *and* time curvature each contribute one half.

FFGFT *explains* this factor of 2 from first structure.

B.2 K3 area invariance [K]

$\tilde{T}\cdot m = 1$ (time curvature) and fractal path elongation (space curvature, Doc. 182) are in FFGFT not alternative mechanisms but **complementary halves**. The K3 rule — area invariance of the time-space 2-cell — anchors this:

$$L_t \cdot l = \text{const.}$$

This is formally the reciprocity form of $\tilde{T} \cdot m = 1$, applied to the time-space pair.

FE verification [K]: Ray tracing through three configurations (30001 log nodes, RK4, `ffgft_308_p36_stufeB1_faktor2_fe.py`):

Configuration	α at solar limb	Ratio to $1.75''$
Time rate only ($\tilde{T} \cdot m = 1$)	$0.875''$	0.500
Path elongation only (space)	$0.875''$	0.500
Both together (K3)	$1.750''$	1.000

PPN test [K]: K3 is the sole survivor of the T^4 selection computation (four candidates tested, `ffgft_308_p36_stufeB2_kappa_herleitung.py`): $\gamma_{\text{PPN}} = 1$ exactly. Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ passed with unlimited margin.

B.3 The same structure in the inertia regime: a_0 [K]/[S]

The K3 two-halves structure appears a second time, at cosmological scale. From Doc. 279 (the exponent 10 is calibrated to H_0, P20/P39; Doc. 309):

$$\frac{H_0}{c} = \frac{\pi}{2}\frac{\xi^{10}}{\bar{\lambda}_e} \qquad \Longrightarrow \qquad R_H = \frac{2\,\bar{\lambda}_e}{\pi\,\xi^{10}}.$$

R_H is the geometric maximal scale of the compact T^4 geometry (Doc. 182) — not an expansion horizon, valid in the static reading.

The cycle doubling [S]: The K3 2-cell has two cycles. A closed path that traverses each once measures:

$$L = \underbrace{2\pi R_H}_{\text{time cycle}} + \underbrace{2\pi R_H}_{\text{space cycle}} = 4\pi R_H.$$

Rindler saturation [S]: The Rindler horizon of an observer with acceleration a lies at c^2/a. When it reaches L, the acceleration is no longer relationally definable. This sets:

$$a_{\text{bg}} = \frac{c^2}{4\pi R_H} = \frac{c^2\,\xi^{10}}{8\,\bar{\lambda}_e}.$$

Inertia from $\tilde{T}\cdot m = 1$ [S]: The foundational relation is a statement about mass. If inertia follows the time field (Unruh coupling), the force law in closed form is:

$$\boxed{a = \sqrt{a_N^2 + a_N a_0}} \qquad \text{with} \quad a_0 = 2\,a_{\text{bg}} = \frac{c^2\,\xi^{10}}{4\,\bar{\lambda}_e} = 1.033\times 10^{-10}\ \text{m/s}^2.$$

The 2π of the geometry cancels exactly against the $\pi/2$ from Doc. 279. The rational factor 4 remains:

$$4 = \underbrace{2}_{\text{K3: time and space cycle}} \times \underbrace{2}_{\text{Unruh: } a_0 = 2a_{\text{bg}}}.$$

No step contains a further free parameter; the scale takes over the calibration of the exponent 10 to H_0.

Chapter C

The Dark Sector

C.1 Dark energy: not a physical object

Λ arises in ΛCDM as an answer to the apparent accelerated expansion. In the static reading this expansion does not exist. Λ thereby loses its referent — it is not falsified by the reading change, it becomes objectless.

Three conditions speak against it:

1. **Only from degenerate data.** SNe Ia, BAO, CMB — exactly the datasets between which both readings are indistinguishable.
2. **No independent anchor.** The only theoretical prediction (QFT vacuum energy) misses the value by 10^{122}.
3. **Objectless under reading change.** Not refuted, but without referent.

That is the status: reading artefact. Not "dark energy does not exist" — that would be a statement within the expansion reading — but: it is a reading-internal bookkeeping quantity.

C.2 Dark matter: a regime, not a substance

The findings — rotation curves, gravitational lensing, Bullet Cluster — are reading-independent. They do not disappear under a reading change. FFGFT traces them, without a new ingredient, to the same force law that $\tilde{T} \cdot m = 1$ together with K3 geometry delivers; this is established for the gas-dominated dwarf galaxy DDO 154, for spiral galaxies and the Bullet Cluster it is open (see below).

Dark matter in FFGFT is not a substance. It is a name for a regime.

The force law in the galactic regime [K]/[S]:

$$a = \sqrt{a_N^2 + a_N a_0}.$$

For $a_N \gg a_0$ (inner regions): $a \to a_N$ (Newton). For $a_N \ll a_0$ (outer regions): $a \to \sqrt{a_N a_0}$, hence $v^4 = GM_b a_0$ — the baryonic Tully-Fisher relation.

In FFGFT, dark matter does not denote a missing mass component but the regime below a_0 in which inertia is non-linear. The missing mass is not missing — inertia is different there.

Results [K] (`ffgft_308_p36_stufeA_a0_anker.py`):

Galaxy	Type	v_{Newton}	v_{FFGFT}	v_{obs}
DDO 154	dwarf, gas-dominated	18.5	47.1	47.0 km/s
NGC 3198	spiral	50.4	121	150 km/s
Milky Way	spiral	122	180	220 km/s

DDO 154 is the cleanest test (gas-dominated, M_b best known): $v_{\mathrm{FFGFT}}/v_{\mathrm{obs}} = 1.002$. The deviations for NGC 3198 and the Milky Way ($121/150 - 1 = -19.3\,\%$, $180/220 - 1 = -18.2\,\%$) lie well outside the M_b uncertainty: for $v \propto M_b^{1/4}$, $\pm 30\,\%$ in M_b correspond to only $+6.8\,\%/-8.5\,\%$ in v (even in the Newtonian limit only $+14\,\%$). Closing them would require $M_b \times 2.36$ (NGC 3198) and $\times 2.23$ (Milky Way). For spirals the test is open.

Bullet Cluster [K] (`ffgft_308_p36_stufeC1_bullet_cluster.py`): Two-clump FE projection (Clowe+2006 parameters) at instantaneous coupling. According to Clowe et al. 2006 the X-ray gas is the dominant baryonic component ($M_{\mathrm{gas}}/M_{\mathrm{stars}} \approx 5$–$10$; gas ~85 % of the baryons, i.e. $5.7 : 1$). With these baryonic masses the lensing peak lies at 240 kpc, i.e. at the gas; it moves to the gas already from gas:stars ≈ 1.4. A galaxy mass of $2.3 \cdot 10^{14}\,M_\odot$ would be the lensing mass including the missing component; a test with it is circular. The Bullet Cluster is therefore not established as passed; the test is open.

Recent data [Q]/[S]: Weak lensing around isolated galaxies shows flat circular velocities, reliably out to at least 300 kpc, and the baryonic Tully-Fisher relation (Mistele et al. 2024) [Q]; this corresponds to the outer limit of the force law. Wide binaries below a_0 are disputed: a deviation from Newton at $4.9\,\sigma$ (Chae et al. 2026) against a model-dependent analysis of the same data that is consistent with Newton (Saad, Ting 2026) [Q]. The force law for systems in the external field of the Milky Way has not been derived in the corpus [S]. For the LZ event at 248 keV and the advance commitment of FFGFT see Doc. 396.

C.3 Asymmetry in comparison

ξ is cross-anchored — it carries the lepton mass ladder, the Koide scalar, the α reference, K_{frak} and k^* (Doc. 275). It works in five foreign sectors before the cosmological one.

Λ works nowhere except its own fit.

That is the difference: not “true vs. false” but “cross-anchored vs. isolated”.

Chapter D

Summary: Picture of the Cosmic Sector

Quantity	FFGFT	Status
Cosmic redshift	mass reading, equivalent to FRW (Doc. 392)	[B]/[K]
Distance law $1+z=e^{\xi x}$	x as coordinate distance: Lyman-α $5.3\,\sigma$	[X]
Achromatic	geometric, FE confirmed	[K]
Degeneracy SNe/BAO/CMB	between both readings	[K]
Λ (DE)	reading artefact, no referent	[K]
Hubble H_0	$c\pi\xi^{10}/(2\bar{\lambda}_e)$, exponent calibrated	[POSIT]
Light deflection	$0.875'' + 0.875'' = 1.750''$	[K]
$\gamma_{\mathrm{PPN}} = 1$	K3 sole survivor	[K]
$a_0 = c^2\xi^{10}/(4\bar{\lambda}_e)$	chain via calibrated H_0	[K]
Rotation curves	DDO 154 ratio 1.002; spirals ~18 % low	[K]; spirals open
Bullet Cluster	baryonic: peak at the gas (240 kpc)	open
K3 cycles $4\pi R_H$	time + space cycle equal	[S]
Rindler saturation	horizon meets maximal scale	[S]
$\xi = 4/30000$	magnitude	[POSIT]
K3 forward proof	2-cell vs. 4-volume	open
14% below a_0^{emp}	RAR $1.20 \pm 0.02\,(\mathrm{rand.}) \pm 0.24\,(\mathrm{syst.}) \times 10^{-10}$: $0.7\,\sigma_{\mathrm{syst}}$, $8\,\sigma_{\mathrm{rand}}$	open

The cosmic sector of FFGFT rests on a single mechanism: $\tilde{T}\cdot m = 1$ together with K3 area invariance produces light deflection, geometric gravity and achromatic redshift. a_0 follows from the same chain without a further free parameter; the scale carries the calibration of H_0.

Dark energy is not a physical object — it is a construct of the expansion reading. Dark matter as particles is not required insofar as the effects follow from the geometry — established for gas-dominated dwarf galaxies, open for spiral galaxies and the Bullet Cluster.

Chapter E

References

Internal documents: Doc. 025 (cosmology overview); Doc. 026 (FE geometry); Doc. 064 (redshift foundations); Doc. 182 (maximal scale); Doc. 267 (cosmological degeneracy); Doc. 279 (H_0 from ξ^{10} chain, K6); Doc. 280 (achromatic redshift); Doc. 275 (k^* recursion); Doc. 190 (P33, P44, K6).

Verification scripts (all inputs declared): `ffgft_308_p36_stufeA_a0_anker.py` (chain $\xi \to a_0$, rotation curves); `ffgft_308_p36_stufeB1_faktor2_fe.py` (factor-2 gate, 30001 log nodes, RK4); `ffgft_308_p36_stufeB2_kappa_herleitung.py` (selection computation K3); `ffgft_308_p36_stufeC1_bullet_cluster.py` (Bullet Cluster projection).

External references: McGaugh, Lelli, Schombert 2016, PRL 117, 201101 (RAR/a_0); Clowe et al. 2006, ApJ 648, L109 (Bullet Cluster); Mistele et al. 2024, arXiv:2406.09685 (weak lensing); Chae et al. 2026, arXiv:2601.21728; Saad, Ting 2026, arXiv:2603.11015 (wide binaries).